ANALYTICAL ABILITY AND LOGICAL REASONING

1. Which number replaces the question mark in the figure given below?

8 | 7	12 | 12	4 | 1	2 | 4	6 | 3
10 | 10	9 | 21	8 | 4	1 | 7	9 | ?

(a)11 (b)6 (c)3 (d)21

2. **Statement I :** Out of total of 200 readers, 100 read Indian Express, 120 read Times of India and 50 read Hindu.
Statement II : Out of total of 200 readers, 100 read Indian Express, 120 reads Times of India and 50 read neither.
How many people (from the group surveyed) read both Indian Express and Times of India?
(a) The Question can be answered with the help of statement II, alone.
(b) Both, statement I and statement II are needed to answer the question.
(c) The question can be answered with the help of statement I alone.
(d) The question cannot be answered even with help of both the statements.

3. If137 + 276 = 435, how much is 731+672=?
(a)534 (b)1403 (c)1623 (d)1531

4. Study the information carefully and answer the questions given below:
If we arrange the alphabets in the word "RATE" in the English alphabetical order, word "AERT" is formed. Then the third alphabet from the left in this word is "R". Similarly, from the word "OPEN" we get-"ENOP" and the third alphabet from the left is "O". From the word "CHEF" we get "CEFH" the third alphabet from the left "F". From the word "TOY" we get-"OTY" and the third alphabet from the left is "Y". From the word "70P" we qu-"DPT" and the third alphabet from the left is 'T'. If we use all these letters, then a meaningful English word "FORTY" can be formed. Now find which of the following word set DOES NOT give a meaningful word in the similar way.
(a) SAME, ROOM, BEST, AUTO
(b) GOAT, PEST, WATT, ARMY
(c) MALE, FIND, LOST, THAT
(d) JUMP, LIME, DUMB, SOME

5. Navjivan Express from Ahmedabad to Chennai leaves Ahmedabad at 6:30 am and travels at 50 kmph towards Baroda situated 100 km away. At 7:00 am Howrah-Ahmedabad and trevels at 40 kmph. At 7:00 am Mr.Shah , the traffic controller at Baroda realizes that both train are running on the same track. How much time does he have to avert a head- on collision between the two train?
(a)15 min (b)20 min
(c)25 min (d)30 min

6. If the points $P(a^2, a)$ lie in the region corresponding to the acute angle between the lines $x = 2y$ *and* $x = 4y$ then
(a)$a \in (2,6)$ (b) $a \in (4,6)$
(c) $a \in (2,4)$ (d) $a \in (10,14)$

7. Some friends planned to contribute equally to jointly buy a CD player. However, two of them decided to withdraw at the last minute. As a result, each of the others had to shell out one rupee more than what they had planned for. If the price (in ₹) of the CD player is an integer between 1000 and 1100, find the number of friends who actually contributed?
(a) 44 (b) 23 (c) 21 (d) 46

8. Two liquids A and B are in the ratio 5:1 in container 1 and in the ratio 1:3 in container 2. In what ratio should the contents of the two containers be mixed so as to obtain a mixture of A and B in the ratio 1:1?
(a)2:3 (b) 4:3 (c)3:2 (d)3:4

9. Each family in a locality has at most adults, and no family has fewer than 3 children. Considering all the families together , there are more adults than boys, more boys than girls, and more girls than families. Then , the minimum possible number of families in the locality is
(a)4 (b)3 (c)2 (d)5

10. Fresh grapes contain 90% by weight while dried grapes contain 20% water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?
(a)2.5 kg (b)2.4 kg (c)2 kg (d)10 kg

Directions (Q. Nos. 61 and 62) for questions 61 and 62 Answer the questions on the basis of the information given below :
A, B, C, D, E and F are a group of friends. There are two housewives, one professor, one engineer, one accountant and one lawyer in the group. There are only two married couples in the group. The lawyer is married to D, who is a housewife. No woman in the group is either an engineer or an accountant. C, the accountant, is married to F, who is a professor. A is married to a housewife. E is not a housewife.

11. What is E's profession?
(a)Accountant (b)Lawyer
(c)Professor (d)Engineer

12. How many members of the group are males?
(a)2 (b)3
(c)4 (d) Cannot be determined

13. Wrong number in the series 7,8,18,57,228,1165,6996 is
(a)288 (b)18 (c)57 (d)28

Directions (Q. Nos. 14 and 15):
Each of the questions given below consists of a statement and/or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the given question. Read both statements and
Give the answer **(U)** if the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

Give the answer **(V)** if the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.
Give the answer **(W)** if the data either in Statement I or in Statement II alone are sufficient to the answer of the question.
Give the answer **(X)** if the data in both statements I and II together are not sufficient to answer the question.
Give the answer **(Y)** if the data in both statements I and II together are necessary to answer the question.

14. How much time will leak take to empty the full cistern?
    I. The cistern is normally filled in 9 h.
    II. It takes one hour more than the usual time to fill the cistern because of a leak in the bottom.
    (a)V (b)U (c)X (d)Y
15. How will long it take to empty the tank if both the inlet pipe P1 and the outlet pipe P2 are opened simultaneously?
    I. P_1 can fill the tank in 16 min.
    II. P_2 can empty the full tank in 8 min.
    (a)X (b)U (c)Y (d)V
16. How many positive numbers less than 10000 are such that the product of their digits is 210?
    (a)36 (b)42 (c)48 (d)54
17. Each of the five people K, L, M, P and Q is of a different weight. It is known that the number of people heavier than P is the same as the number of people lighter than Q. L is the heaviest and K is not the lightest. Who is the lightest?
    (a)M (b)L (c)Q (d)P
18. John, Johny and Janardan participated in a race and each won a different medal among Gold, Silver and Bronze, not necessarily in that order. Each person among them gives two replies to any question, one of which is true and the other is false (in any order). When asked about the details of the medals obtained by them, the following were their replies.
    John: I won the Gold medal. Johny won the Bronze medal.
    Johny : John won the Silver medal. I won the Gold medal.
    Janardan: Johny won the Silver medal. I won the Gold medal.
    Which among the following is the correct order of the persons who won the Gold medal, the Silver medal and the Bronze medal respectively?
    (a)John, Johny, Janardan (b)Janardan, John, Johny
    (c)Johny, Janardan, John (d)Janardan, Johny, John
19. Each of A, B and C is a different digit among 1 to 9. How many different values of the sum of A. B and C are possible, if $ABA \times AA = ACCA$?
    (a)1 (b)3 (c) 7 (d)8
20. In a certain language, if the word 'BASKET is coded as 'UFLTBC', then how is the word 'SIMPLE' coded in that language?
    (a)FMQNJT (b)FMQGNJ
    (c)FMQNJH (d)MFNQTJ
21. A dealer offers sells half of the eggs that he has and another half an egg to Anurag. Then, he sells half of the balance eggs and another half an egg to Deepak. Then, he sells half of the balance eggs and another half an egg to Sivani. In the end he is left with just 7 eggs and he claims that he never broke an egg. How 'many eggs did he start with?
    (a)65 (b)63 (c)67 (d)69
22. There are eight poet, namely, P, Q, R, S, T, U, V and W and H in respect of whom questions are being asked in the examination. The first four are ancient poets and the last four are modern poets. The question on ancient and modern poets is being asked in alternate years. Those who like W also like V, those who like S like R also. The examiner who sets question is not likely to ask question on S because he has written an article on him. But he likes S. Last year a question was asked on U. Considering these facts, on whom the question is most likely to be asked this year?
    (a)Q (b)R (c)S (d)V
23. A team must be selected from ten probable's-A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards, B, G and H are point guards and D, F and I are defenders. The team must have at least one forward,one point guard and one defender. If the team includes J, it must also include F. The team must include E or B, but not both. If the team includes G, it must also include F. The team must include exactly one among C, G and I. C and F cannot be members of the same team. D and H cannot be members of the same team. The team must include both A and D or neither of them. There is no restriction on the number of members in the team. What could be the maximum size of the team that includes G?
    (a)4 (b)5
    (c)6 (d)More than 6
24. How many 4-digit numbers that can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?
    (a)216 (b)60 (c)24 (d)25
25. In a family of six persons, there are people from three generations. Each person has separate profession and also they like different colours. There are two couples in the family. Rohan is a CA and his wife neither is a doctor nor likes green colour. Engineer likes red colour and his wife is a teacher. Mohini is mother in law of Savita and she likes orange colour. Deepak is grandfather of Titu and Titu, who is a principal, likes black colour. Nerru is granddaughter of Mohini and she likes blue colour. Nerru's mother likes white colour. Savita is a
    (a)Doctor (b)Teacher
    (c)Housewife (d)None of these
26. 1. All chickens are birds.
    2. Some chickens are hens.
    3. Female birds lay eggs.

If the above three statements are facts,which of the following statement must also be a fact?

I. All birds lay eggs.
II. Hens are birds.
III. Some chickens are not hens.

(a) II Only
(b) II and III only
(c) I, II and III
(d) None of the statements is known as fact

27. The number that comes next in the series 1,2,3,6,11,20,37,68,....
(a)105 (b)124 (c)125 (d)126

28. Using only 2, 5, 10, 25 and 50 paise coins, the smallest number of coins required to pay exactly 79 paise, 66 and ₹1.01 to three different persons is
(a) 17 (b) 20 (c) 19 (d) 18

29. What comes next in the following sequence
99, 90, 83, 78, 75,
(a)74 (b)69 (c) 67 (d)63

30. A dealer offers a cash discount of 20% and still makes a profit of 20%, when he further allows 16 articles to a dozen to a particularly sticky bargainer. How much per cent above the actual price were his was the listed price of the article?
(a)100% (b)80% (c)75% (d)66%

Questions (Q. Nos. 31-33) are based on the following instructions,
Twelve classmates A, B, C, D, E, F, G, H, I, J, K and L are sitting on a square table with 3 persons on each side. ABC and GJK are sitting on opposite sides. A and L are adjacent to each other. K is sitting diagonally to C.

31. If F is sitting between D and E, who is sitting to the left of K?
(a)H (b)I (c)H or I (d)None

32. If H is sitting between L and F, then who will be facing H?
(a)D (b)E (c) G (d) I

33. If G and E are facing C and L respectively, the neighbours of C are
(a)J and H (b)J and E
(c)H and J (d)B and E

34. The integers 34041 and 32506, when divided by a 3- digit integer n, leave the same remainder. What can be the value of n?
(a)289 (b)307 (c)367 (d)493

35. The number of solid spheres, each of diameter 3 cm that could be moulded to form a solid metal cylinder of height 54 cm and diameter 4 cm is
(a)16 (b)24 (c)36 (d)48

36. A clock is set right at 5 AM. The clock loses 16 min in 24 h. What will be the true time when the clock indicates 10 PM on 4th day?
(a)11:15 PM (b)11:00 PM
(c)12:00 PM (d)12:30 PM

37. A train overtakes two persons who are walking in the same direction in which the train is going, at the rate of 2 kmph and 4 kmph and passes them completely in 9 sec and 10 sec respectively. The length of the train is
(a)72 m (b)54 m (c)50 m (d)45 m

38. Decide which of the give conclusions logically follow from the given students(s)
Statements:
All suns are moons.
Some moons are planets
Conclusions:
I. All moons are suns.
II. At least some moons are planets.
(a) Either conclusion I or II is true
(b) Neither conclusion I nor II is true
(c) Both conclusion I or II is true
(d) Only conclusion II is true

39. Ten points are marked on a straight line and eleven points are marked on another straight line. How many triangles can be constructed with vertices from among the above points?
(a)495 (b)550 (c)1045 (d)2475

40. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is
(a)123 (b)127 (c)235 (d)305

ANSWER

1.	C	2.	A	3.	A	4.	D	5.	B
6.	C	7.	A	8.	D	9.	B	10.	A
11.	D	12.	B	13.	A	14.	D	15.	C
16.	D	17.	A	18.	B	19.	D	20.	A
21.	B	22.	B	23.	C	24.	B	25.	D
26.	D	27.	C	28.	D	29.	A	30.	A
31.	C	32.	D	33.	D	34.	B	35.	D
36.	B	37.	C	38.	D	39.	C	40.	B